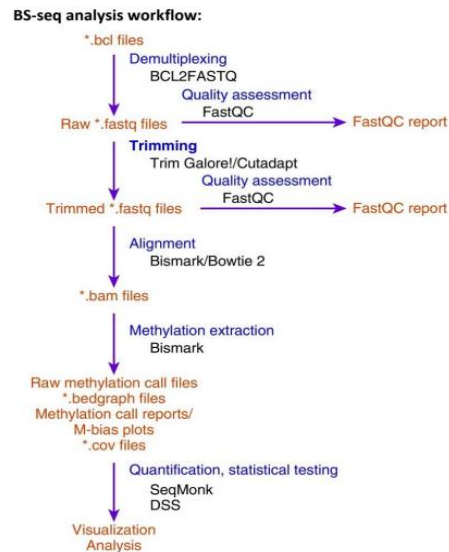


Assignment-16

245HSBD016 (Yash Priya Baid)

Goal: To perform BS-seq analysis using Bismark.



Step-1: Download the raw sequencing data (FASTQ files) for the two samples (SRR6380917 and SRR6380922) from the ENA database.

Organism: [Homo sapiens \(human\)](#)

Instrument Platform: ION_TORRENT

Instrument Model: Ion Torrent PGM

Read Count: 1163

Base Count: 194483

Center Name: SUB3347979

Library Layout: SINGLE

Library Strategy: Bisulfite-Seq

Library Source: OTHER

Library Name: IonX83

Show More

Read Files

Show Column Selection

Download report: JSON TSV

Get download script

Download selected files

Study Accession	Sample Accession	Experiment Accession	Run Accession	Tax Id	Scientific Name	Generated FASTQ files: FTP	Submit F
PRJNA422608	SAMN08182928	SRX3474950	SRR6380917	9606	Homo sapiens	☐ SRR6380917.fastq.gz	☐

Run: SRR6380922

Ion Torrent PGM sequencing. Targeted BS-seq of full-length control human mitochondrial DNA; 100% methylation

Organism: [Homo sapiens \(human\)](#)

Instrument Platform: ION_TORRENT

Instrument Model: Ion Torrent PGM

Read Count: 11462

Base Count: 1732119

Center Name: SUB3347979

Library Layout: SINGLE

Library Strategy: Bisulfite-Seq

Library Source: OTHER

Library Name: IonX91

Show More

Read Files

Show Column Selection

Download report: JSON TSV

Get download script

Download selected files

Study Accession	Sample Accession	Experiment Accession	Run Accession	Tax Id	Scientific Name	Generated FASTQ files: FTP	Submit F
PRJNA422608	SAMN08182933	SRX3474945	SRR6380922	9606	Homo sapiens	☐ SRR6380922.fastq.gz	☐

Step-2: Create these sub-directories: 1_Raw_data, 2_Preprocessing, 3_Reference, 4_Alignment, 5_Methylation

Step-3: Move the downloaded .fastq.gz files to the 1_Raw_data folder using the mv command.

```
(base) ibab@IBAB-MScBDB2-Comp007:~$ cd NGS
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS$ mkdir BS_Seq
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq$ mkdir 1_Raw_data 2_Preprocessing 3_Reference 4_Alignment 5_Methylation
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq$ ls
1_Raw_data 2_Preprocessing 3_Reference 4_Alignment 5_Methylation
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq$ cd
(base) ibab@IBAB-MScBDB2-Comp007:~$ cd Downloads/
(base) ibab@IBAB-MScBDB2-Comp007:~/Downloads$ mv SRR6380917.fastq.gz SRR6380922.fastq.gz ~/NGS/BS_Seq/1_Raw_data/

(base) ibab@IBAB-MScBDB2-Comp007:~/Downloads$ cd ..
(base) ibab@IBAB-MScBDB2-Comp007:~$ cd NGS
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS$ cd BS_Seq/
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq$ cd 1_Raw_data/
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/1_Raw_data$ ls
SRR6380917.fastq.gz SRR6380922.fastq.gz
```

Step-4: Check the quality of the FASTQ files using FastQC.

`fastqc /home/ibab/NGS/BS_Seq/1_Raw_data/*.gz -o .`

SRR6380917

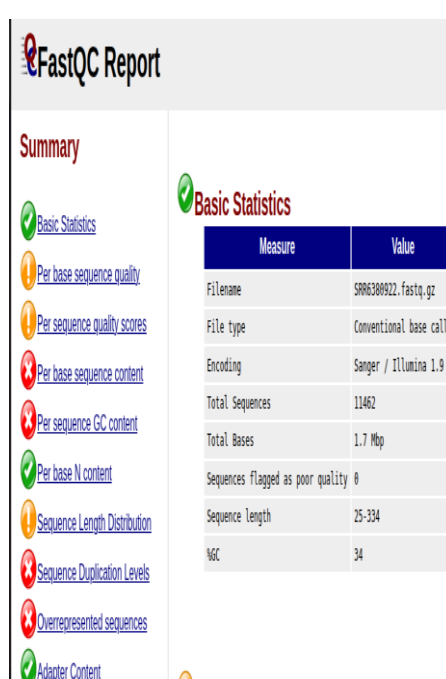


FastQC Report for SRR6380917.fastq.gz. The report shows a summary of basic statistics and a list of checks. The basic statistics table is as follows:

Measure	Value
Filename	SRR6380917.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	1163
Total Bases	194.4 kbp
Sequences flagged as poor quality	0
Sequence length	26-242
%GC	37

The checks on the left include: Basic Statistics (pass), Per base sequence quality (pass), Per sequence quality scores (pass), Per base sequence content (fail), Per sequence GC content (fail), Per base N content (pass), Sequence Length Distribution (pass), Sequence Duplication Levels (fail), Overrepresented sequences (fail), and Adapter Content (pass).

SRR6380922



FastQC Report for SRR6380922.fastq.gz. The report shows a summary of basic statistics and a list of checks. The basic statistics table is as follows:

Measure	Value
Filename	SRR6380922.fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	11462
Total Bases	1.7 Mbp
Sequences flagged as poor quality	0
Sequence length	25-334
%GC	34

The checks on the left include: Basic Statistics (pass), Per base sequence quality (pass), Per sequence quality scores (pass), Per base sequence content (fail), Per sequence GC content (fail), Per base N content (pass), Sequence Length Distribution (pass), Sequence Duplication Levels (fail), Overrepresented sequences (fail), and Adapter Content (pass).

Step-5: Then, remove the low-quality data and adapters using Trim Galore!

`trim_galore --fastqc --gzip /home/ibab/NGS/BS_Seq/1_Raw_data/*.fastq.gz -o .`

SRR6380917

SRR6380922

FastQC Report	
Summary	
Basic Statistics	
Per base sequence quality	
Per sequence quality scores	
Per base sequence content	
Per sequence GC content	
Per base N content	
Sequence Length Distribution	
Sequence Duplication Levels	
Overrepresented sequences	
Adapter Content	

FastQC Report	
Summary	
Basic Statistics	
Per base sequence quality	
Per sequence quality scores	
Per base sequence content	
Per sequence GC content	
Per base N content	
Sequence Length Distribution	
Sequence Duplication Levels	
Overrepresented sequences	
Adapter Content	

Step-6: Since, we already have downloaded human reference genome FASTA file (hg38), so just move this to 3_Reference folder using mv command.

Step-7: Use the hg38 reference genome with Bismark to create two bisulfite-converted versions of the genome (C-to-T and G-to-A). Bismark then uses the Bowtie 2 indexing tool to build the necessary alignment indices for both converted genomes.

Since indexing the whole reference genome will take a long time, we will only extract the data for Chromosome 15 and then run the Bismark genome indexing.

`/home/ibab/NGS_Lab/ngs_softwares/sam/samtools faidx hg38.fa chr15 > chr15.fasta`

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/3_Reference$ /home/ibab/NGS_Lab/ngs_softwares/sam/samtools faidx hg38.fa chr15 > chr15.fasta
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/3_Reference$ ls -lh
total 3.3G
-rw-rw-r-- 1 ibab ibab 100M Oct  9 14:52 chr15.fasta
-rw-rw-r-- 1 ibab ibab 3.2G Oct 28  2022 hg38.fa
-rw-rw-r-- 1 ibab ibab 31K Oct  9 14:49 hg38.fa.fai
```

`bismark_genome_preparation --bowtie2 .`

Note: The above command automatically handles indexing internally. So, no need to run samtools faidx on chr15.fasta again.

Also, a separate directory was needed because running bismark_genome_preparation in the main folder (3_Reference folder) could mix index files from chr15.fasta and the whole hg38.fa, causing confusion. Keeping chr15.fasta in its own folder (chr15_index folder) ensures that only its bisulfite indexes are generated and used.

```
chr15.fasta NGS3.fa NGS3.fa.fast
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/3_Reference$ mkdir chr15_index
mv chr15.fasta chr15_index/
cd chr15_index
bismark_genome_preparation --bowtie2 .
Writing bisulfite genomes out into a single MFA (multi FastA) file

Bisulfite Genome Indexer version v0.25.1 (last modified: 19 May 2022)

Step I - Prepare genome folders - completed
```

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/3_Reference/chr15_index$ ls -lh
total 100M
drwxrwxr-x 4 ibab ibab 4.0K Oct  9 15:05 Bisulfite_Genome
-rw-rw-r-- 1 ibab ibab 100M Oct  9 14:52 chr15.fasta
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/3_Reference/chr15_index$ cd Bisulfite_Genome/
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/3_Reference/chr15_index/Bisulfite_Genome$ ls
CT_conversion  GA_conversion
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/3_Reference/chr15_index/Bisulfite_Genome$ cd CT_conversion/
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/3_Reference/chr15_index/Bisulfite_Genome/CT_conversion$ ls -lh
total 222M
-rw-rw-r-- 1 ibab ibab 31M Oct  9 15:06 BS_CT.1.bt2
-rw-rw-r-- 1 ibab ibab 21M Oct  9 15:06 BS_CT.2.bt2
-rw-rw-r-- 1 ibab ibab 179 Oct  9 15:06 BS_CT.3.bt2
-rw-rw-r-- 1 ibab ibab 21M Oct  9 15:06 BS_CT.4.bt2
-rw-rw-r-- 1 ibab ibab 31M Oct  9 15:07 BS_CT.rev.1.bt2
-rw-rw-r-- 1 ibab ibab 21M Oct  9 15:07 BS_CT.rev.2.bt2
-rw-rw-r-- 1 ibab ibab 100M Oct  9 15:06 genome_mfa.CT_conversion.fa
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/3_Reference/chr15_index/Bisulfite_Genome/CT_conversion$ cd ..
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/3_Reference/chr15_index/Bisulfite_Genome$ cd GA_conversion/
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/3_Reference/chr15_index/Bisulfite_Genome/GA_conversion$ ls -lh
total 222M
-rw-rw-r-- 1 ibab ibab 31M Oct  9 15:06 BS_GA.1.bt2
-rw-rw-r-- 1 ibab ibab 21M Oct  9 15:06 BS_GA.2.bt2
-rw-rw-r-- 1 ibab ibab 179 Oct  9 15:06 BS_GA.3.bt2
-rw-rw-r-- 1 ibab ibab 21M Oct  9 15:06 BS_GA.4.bt2
-rw-rw-r-- 1 ibab ibab 31M Oct  9 15:07 BS_GA.rev.1.bt2
-rw-rw-r-- 1 ibab ibab 21M Oct  9 15:07 BS_GA.rev.2.bt2
-rw-rw-r-- 1 ibab ibab 100M Oct  9 15:06 genome_mfa.GA_conversion.fa
```

Step-8: Use the Bowtie 2 aligner to map the single-end sequencing reads from the trimmed FASTQ file against the bisulfite-converted genome indices located in the reference folder. Finally, the BAM file was sorted and indexed using Samtools.

```
bismark --bowtie2 -p 3 -o ~/NGS/BS_Seq/4_Alignment/
~/NGS/BS_Seq/3_Reference/chr15_index/
~/NGS/BS_Seq/2_Preprocessing/SRR6380917_trimmed.fq.gz
```

```
bismark --bowtie2 -p 3 -o ~/NGS/BS_Seq/4_Alignment/
~/NGS/BS_Seq/3_Reference/chr15_index/
~/NGS/BS_Seq/2_Preprocessing/SRR6380922_trimmed.fq.gz
```

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/4_Alignment$ ls -lh
total 16K
-rw-rw-r-- 1 ibab ibab 447 Oct  9 15:25 SRR6380917_trimmed_bismark_bt2.bam
-rw-rw-r-- 1 ibab ibab 1.8K Oct  9 15:25 SRR6380917_trimmed_bismark_bt2_SE_report.txt
-rw-rw-r-- 1 ibab ibab 938 Oct  9 15:36 SRR6380922_trimmed_bismark_bt2.bam
-rw-rw-r-- 1 ibab ibab 1.8K Oct  9 15:36 SRR6380922_trimmed_bismark_bt2_SE_report.txt
```

SRR6380917

SRR6380922

<pre> Final Alignment report ===== Sequences analysed in total: 11322 Number of alignments with a unique best hit from the different alignments: 5 Mapping efficiency: 0.0% Sequences with no alignments under any condition: 11310 Sequences did not map uniquely: 7 Sequences which were discarded because genomic sequence could not be extracted: 0 Number of sequences with unique best (first) alignment came from the bowtie output: CT/CT: 2 ((converted) top strand) CT/GA: 3 ((converted) bottom strand) GA/CT: 0 (complementary to (converted) top strand) GA/GA: 0 (complementary to (converted) bottom strand) Number of alignments to (merely theoretical) complementary strands being rejected in total: 0 Final Cytosine Methylation Report ===== Total number of C's analysed: 68 Total methylated C's in CpG context: 7 Total methylated C's in CHG context: 9 Total methylated C's in CHH context: 37 Total methylated C's in Unknown context: 0 Total unmethylated C's in CpG context: 0 Total unmethylated C's in CHG context: 0 Total unmethylated C's in CHH context: 15 Total unmethylated C's in Unknown context: 0 C methylated in CpG context: 100.0% C methylated in CHG context: 100.0% C methylated in CHH context: 71.2% Can't determine percentage of methylated Cs in Unknown context (CN or CHN) if value was 0 </pre>	<pre> Final Alignment report ===== Sequences analysed in total: 1159 Number of alignments with a unique best hit from the different alignments: 1 Mapping efficiency: 0.1% Sequences with no alignments under any condition: 1158 Sequences did not map uniquely: 0 Sequences which were discarded because genomic sequence could not be extracted: 0 Number of sequences with unique best (first) alignment came from the bowtie output: CT/CT: 1 ((converted) top strand) CT/GA: 0 ((converted) bottom strand) GA/CT: 0 (complementary to (converted) top strand) GA/GA: 0 (complementary to (converted) bottom strand) Number of alignments to (merely theoretical) complementary strands being rejected in total: 0 Final Cytosine Methylation Report ===== Total number of C's analysed: 5 Total methylated C's in CpG context: 0 Total methylated C's in CHG context: 1 Total methylated C's in CHH context: 4 Total methylated C's in Unknown context: 0 Total unmethylated C's in CpG context: 0 Total unmethylated C's in CHG context: 0 Total unmethylated C's in CHH context: 0 Total unmethylated C's in Unknown context: 0 Can't determine percentage of methylated Cs in CpG context if value was 0 C methylated in CHG context: 100.0% C methylated in CHH context: 100.0% Can't determine percentage of methylated Cs in Unknown context (CN or CHN) if value was 0 </pre>
--	--

`samtools sort SRR6380917_trimmed_bismark_bt2.bam -o SRR6380917_sorted.bam`

`samtools sort SRR6380922_trimmed_bismark_bt2.bam -o SRR6380922_sorted.bam`

`samtools index SRR6380917_sorted.bam`

`samtools index SRR6380922_sorted.bam`

Step-9: Now, use the Bismark methylation extractor tool on the sorted BAM file to determine the methylation status of every cytosine. The key options used are `--cytosine_report`, which generates a comprehensive, genome-wide report of all cytosines (covered or not) using the specified reference genome.

`bismark_methylation_extractor --cytosine_report --Cx --genome_folder /home/ibab/NGS/BS_Seq/3_Reference /home/ibab/NGS/BS_Seq/4_Alignment/SRR6380917_sorted.bam`

`bismark_methylation_extractor --cytosine_report --Cx --genome_folder /home/ibab/NGS/BS_Seq/3_Reference /home/ibab/NGS/BS_Seq/4_Alignment/SRR6380922_sorted.bam`

The command mentioned above generates multiple output files. The raw methylation calls at the single-read level are contained in the six files ending in `OT.txt` and `OB.txt`, which correspond to CpG, CHG, and CHH contexts. These raw calls combine to produce imp. summary formats -

CpG_report.txt file is the most significant for subsequent statistical analyses, as it presents a normalized matrix for differential methylation analysis, listing every cytosine in the genome along with its position and the total counts of both methylated and unmethylated cytosines (including instances where there are none). In contrast, the .bismark.cov.gz file only provides aggregated counts for cytosines that are covered. Meanwhile, the .bedGraph.gz file offers position-specific methylation percentages, formatted for visualization in genome browsers, thereby representing this aggregated data visually. The cytosine_context_summary.txt and splitting_report.txt files offer a numerical summary of the sample's overall methylation levels and bisulfite conversion efficiency.

Note: The --Cx parameter reports methylation for all cytosine contexts (CpG, CHG, and CHH). By default, Bismark focuses on CpG sites. We use --Cx when some cytosine reports or bedGraph files are missing or when a comprehensive genome-wide methylation profile, including non-CpG contexts, is needed. This ensures all cytosines are included in the output, allowing generation of complete .bedGraph files and cytosine reports.

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/5_Methylation$ ls
CHG_OT_SRR6380922_sorted.txt  CHH_OT_SRR6380922_sorted.txt  SRR6380917_sorted.M-bias.txt      SRR6380922_sorted.M-bias.txt
CHG_OT_SRR6380917_sorted.txt  CpG_OT_SRR6380922_sorted.txt  SRR6380917_sorted_splitting_report.txt  SRR6380922_sorted_splitting_report.txt
CHH_OT_SRR6380922_sorted.txt  SRR6380917_sorted.bedGraph.gz  SRR6380922_sorted.bedGraph.gz
CHH_OT_SRR6380917_sorted.txt  SRR6380917_sorted.bismark.cov.gz  SRR6380922_sorted.bismark.cov.gz
```

SRR6380917

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/5_Methylation$ cat SRR6380917_sorted_splitting_report.txt
SRR6380917_sorted.bam

Parameters used to extract methylation information:
Bismark Extractor Version: v0.25.1
Bismark result file: single-end (SAM format)
Output specified: strand-specific (default)

Processed 1 lines in total
Total number of methylation call strings processed: 1

Final Cytosine Methylation Report
=====
Total number of C's analysed: 5

Total methylated C's in CpG context: 0
Total methylated C's in CHG context: 1
Total methylated C's in CHH context: 4

Total C to T conversions in CpG context: 0
Total C to T conversions in CHG context: 0
Total C to T conversions in CHH context: 0

Can't determine percentage of methylated Cs in CpG context if value was 0
C methylated in CHG context: 100.0%
C methylated in CHH context: 100.0%

(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/5_Methylation$ cat CHG_OT_SRR6380917_sorted.txt
Bismark methylation extractor version v0.25.1
SRR6380917.943_943/1 + chr15 100479131 X

(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/5_Methylation$ cat CHH_OT_SRR6380917_sorted.txt
Bismark methylation extractor version v0.25.1
SRR6380917.943_943/1 + chr15 100479126 H
SRR6380917.943_943/1 + chr15 100479135 H
SRR6380917.943_943/1 + chr15 100479136 H
SRR6380917.943_943/1 + chr15 100479137 H
```

```
(base) lbab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/5_Methylation$ cat SRR6380917_sorted.M-bias.txt
```

CpG context

position	count methylated	count unmethylated	% methylation	coverage
1	0	0		
2	0	0		
3	0	0		
4	0	0		
5	0	0		
6	0	0		
7	0	0		
8	0	0		
9	0	0		
10	0	0		
11	0	0		
12	0	0		
13	0	0		
14	0	0		
15	0	0		
16	0	0		

CHG context

position	count methylated	count unmethylated	% methylation	coverage
1	0	0		
2	0	0		
3	0	0		
4	0	0		
5	0	0		
6	0	0		
7	0	0		
8	0	0		
9	0	0		
10	1	0	100.00	1
11	0	0		
12	0	0		
13	0	0		
14	0	0		
15	0	0		
16	0	0		

CHH context

position	count methylated	count unmethylated	% methylation	coverage
1	0	0		
2	0	0		
3	0	0		
4	0	0		
5	1	0	100.00	1
6	0	0		
7	0	0		
8	0	0		
9	0	0		
10	0	0		
11	0	0		
12	0	0		
13	0	0		
14	1	0	100.00	1
15	1	0	100.00	1
16	1	0	100.00	1

Interpretation (SRR6380917): The BS-seq reads for sample SRR6380917 were aligned to the bisulfite-converted genome and processed using the Bismark Methylation Extractor with strand-specific output. From the splitting report, a total of 5 cytosines were analyzed: 0 methylated in CpG, 1 in CHG, and 4 in CHH context, corresponding to 100% methylation for CHG and CHH cytosines. M-bias analysis shows that methylation occurs at specific positions in CHG and CHH contexts, while no methylation was detected in CpG positions.

SRR6380922

```
(base) lbab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/5_Methylation$ cat CpG_OB_SRR6380922_sorted.txt
```

Bismark methylation extractor version v0.25.1

SRR6380922.7427_7427/1	+	chr15	45023345	Z
SRR6380922.7427_7427/1	+	chr15	45023334	Z
SRR6380922.7427_7427/1	+	chr15	45023329	Z
SRR6380922.7427_7427/1	+	chr15	45023295	Z
SRR6380922.7427_7427/1	+	chr15	45023292	Z
SRR6380922.7427_7427/1	+	chr15	45023289	Z
SRR6380922.7427_7427/1	+	chr15	45023260	Z


```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/5_Methylation$ cat CHG_OB_SRR6380922_sorted.txt
Bismark methylation extractor version v0.25.1
SRR6380922.7427_7427/1 + chr15 45023357 X
SRR6380922.7427_7427/1 + chr15 45023349 X
SRR6380922.7427_7427/1 + chr15 45023335 X
SRR6380922.7427_7427/1 + chr15 45023330 X
SRR6380922.7427_7427/1 + chr15 45023319 X
SRR6380922.7427_7427/1 + chr15 45023296 X
SRR6380922.7427_7427/1 + chr15 45023293 X
SRR6380922.7427_7427/1 + chr15 45023290 X
SRR6380922.7427_7427/1 + chr15 45023268 X
```

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/5_Methylation$ cat CHH_OB_SRR6380922_sorted.txt
Bismark methylation extractor version v0.25.1
SRR6380922.7427_7427/1 + chr15 45023367 H
SRR6380922.7427_7427/1 + chr15 45023366 H
SRR6380922.7427_7427/1 + chr15 45023365 H
SRR6380922.7427_7427/1 + chr15 45023363 H
SRR6380922.7427_7427/1 + chr15 45023362 H
SRR6380922.7427_7427/1 + chr15 45023359 H
SRR6380922.7427_7427/1 + chr15 45023358 H
SRR6380922.7427_7427/1 + chr15 45023354 H
SRR6380922.7427_7427/1 + chr15 45023350 H
SRR6380922.7427_7427/1 + chr15 45023343 H
SRR6380922.7427_7427/1 + chr15 45023338 H
SRR6380922.7427_7427/1 + chr15 45023337 H
SRR6380922.7427_7427/1 + chr15 45023336 H
SRR6380922.7427_7427/1 + chr15 45023325 H
SRR6380922.7427_7427/1 + chr15 45023323 H
SRR6380922.7427_7427/1 + chr15 45023322 H
SRR6380922.7427_7427/1 + chr15 45023320 H
SRR6380922.7427_7427/1 + chr15 45023301 H
SRR6380922.7427_7427/1 + chr15 45023287 H
SRR6380922.7427_7427/1 + chr15 45023286 H
SRR6380922.7427_7427/1 + chr15 45023279 H
SRR6380922.7427_7427/1 + chr15 45023277 H
SRR6380922.7427_7427/1 + chr15 45023270 H
SRR6380922.10970_10970/1 + chr15 49715434 H
SRR6380922.10970_10970/1 - chr15 49715433 h
SRR6380922.10970_10970/1 + chr15 49715431 H
SRR6380922.10970_10970/1 - chr15 49715428 h
SRR6380922.10970_10970/1 + chr15 49715427 H
SRR6380922.10970_10970/1 - chr15 49715424 h
SRR6380922.10970_10970/1 + chr15 49715423 H
SRR6380922.10970_10970/1 - chr15 49715421 h
SRR6380922.10970_10970/1 - chr15 49715420 h
```

Interpretation (SRR6380922): For this sample, cytosine methylation was observed across all sequence contexts. A total of 7 CpG, 9 CHG, and 36 CHH cytosines were identified as methylated. Both CpG and CHG sites showed complete (100%) methylation, while CHH sites exhibited variable methylation levels ranging from 33% to 100%. These results indicate strong methylation activity across both CpG and non-CpG contexts, suggesting widespread cytosine methylation in sample.

```
(base) ibab@IBAB-MScBDB2-Comp007:~/NGS/BS_Seq/5_Methylation$ awk '/CHH context/{f=1;next}/^$/ {f=0} f && ($4==100 || $4==50 || $4==33 || $4==0){c[$4]++} END{for(i in c)print i, c[i]}' SRR6380922_sorted.M-bias.txt
0.00 7
50.00 3
100.00 27
0 70
```

Step-10: Load the generated .bedGraph.gz files for both samples along with chr15.fasta (reference genome) into the IGV Genome Browser.



Step-11: Compare methylation tracks between the two samples and interpret the differences.

Interpretation: When comparing the methylation profiles across chromosome 15, clear differences are visible between the two samples. The SRR6380922 sample shows multiple discrete methylation peaks, while SRR6380917 exhibits an almost flat signal with no notable peaks, indicating minimal methylation activity. Specifically, methylation signals in SRR6380922 are observed at approximately:

- ~40-60 Mb — three distinct peaks in close proximity
- ~90 Mb — one prominent peak

No corresponding peaks are detected in SRR6380917 at these loci, suggesting that these CpG regions are uniquely methylated in SRR6380922. This pattern implies that SRR6380922 is hypermethylated at specific regions of chromosome 15 compared to SRR6380917. Such locus-specific hypermethylation could indicate epigenetic regulation of nearby genes, potentially leading to transcriptional silencing or other chromatin-associated effects in those regions.